

# Group 27 Course Project proposal

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## Abstract

Chain-of-Thought (CoT) prompting can improve the performance of large language models on complex reasoning tasks, but it remains unclear whether the generated reasoning steps faithfully reflect the model's internal reasoning process. This project aims to reproduce the paper *Measuring Chain of Thought Faithfulness by Unlearning Reasoning Steps* and to introduce certain innovations and extensions based on this work.

## 1 Problem

The central research question is whether the CoT (Wei et al., 2022) steps generated by a language model are truly used by the model to produce its final prediction. (Tutek et al., 2025) More specifically, if we remove the information contained in one CoT step from the model parameters, will the model still make the same prediction under a no-CoT prompt? If the prediction changes after a successful and precise unlearning intervention, the removed step can be interpreted as parametrically faithful to the model's internal reasoning. If the prediction does not change, the step may be non-essential, the model may have alternative reasoning paths, or the CoT may be unfaithful.

The paper proposes the **Parametric Faithfulness Framework (PFF)** and **Faithfulness by Unlearning Reasoning steps (FUR)** as a systematic approach to measuring CoT faithfulness. We will comprehensively reproduce the paper and then further extend the method in the following directions:

- Introduce faithfulness-enhancing methods such as **FDORO** (Paul et al., 2024) under this framework to improve faithfulness.
- Address computational efficiency issues when applying the method to large-scale samples.
- Extend the method to the **Tree of Thought** (Yao et al., 2023) setting.

## 2 Dataset

The original paper uses five multi-hop multiple-choice question answering datasets: **OpenBookQA**, **ARC-Challenge**, **StrategyQA**, **TruthfulQA**, and **BigBench-Hard Sports Understanding**. We will reproduce the original results on these datasets. In addition, we will use **C-Eval** (Huang et al., 2023), a Chinese multiple-choice question answering dataset, to extend the authors' results to the Chinese setting with Qwen models. Additionally, we aim to scale the experiments to more examples or more datasets, such as **GSM-8K** (Cobbe et al., 2021). These datasets are well suited for measuring Chain-of-Thought (CoT) faithfulness because they require multi-step reasoning and provide a fixed set of candidate answers.

## 3 Approach

The main method of the entire project is **machine unlearning**. Specifically, we use carefully designed prompts to elicit CoT reasoning from large language models, apply certain post-processing steps to the generated CoT, and then fine-tune the model. The detailed methods include approaches such as **NPO-KL**. The overall workflow of the project is as follows:

- We reproduce the results of the original paper, using models such as LLaMA-3B.
- We plan to extend the results of the paper to Chinese and Chinese large language models.
- We seek to consider extending related methods for improving model faithfulness to the FUR framework (such as the FRODO method)
- Regarding the instance by instance fine-tuning approach used in the paper, we will explore whether fine-tuning can be performed in larger batches to improve efficiency.
- We are also interested in the broader line of research. We will be made to conduct further investigations based on this series of papers.

## References

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## A Appendix

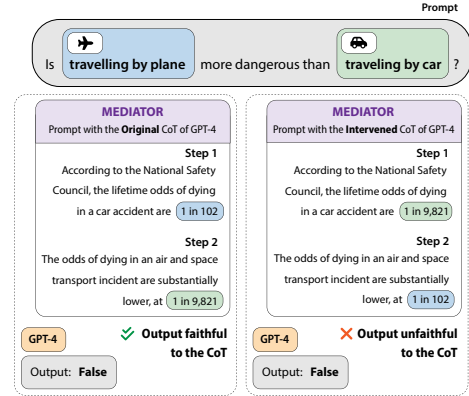


Figure 1: an simple example

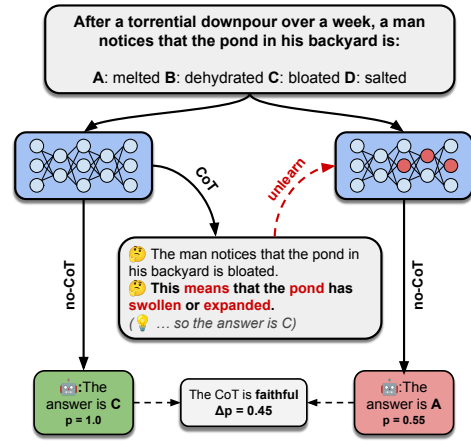


Figure 2: What is faithfulness

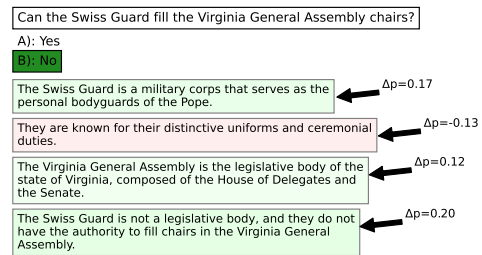


Figure 3: the faithfulness example